

- 7 The decay of radioactive materials is a random process. On average, nuclides which decay rapidly exist for a shorter time than nuclides which decay slowly. It is common practice when making calculations on decay to make use of the half-life of a nuclide. One difficulty that arises with these calculations is when the radioactive material is a mixture of two or more nuclides.

This question considers the case when a mixture of two radioactive nuclides is present. In decommissioning a nuclear power station, this difficulty is compounded by the presence of about a hundred different radioactive nuclides in significant quantities.

- (a) Explain what is meant by a *random* process.

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- (b) Define *half-life*.

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..... [1]

- (c) $^{131}_{52}\text{Te}$ decays by β emission to $^{131}_{53}\text{I}$. $^{131}_{53}\text{I}$ is not stable, and decays by β emission to the stable isotope Xe, but the half-life for this decay is very much longer than that for the decay of $^{131}_{52}\text{Te}$. A sample of pure $^{131}_{52}\text{Te}$ is prepared at time $t = 0$, and Fig. 7.1 shows the variation with time of the activity A of the whole sample. Background radiation can be ignored.

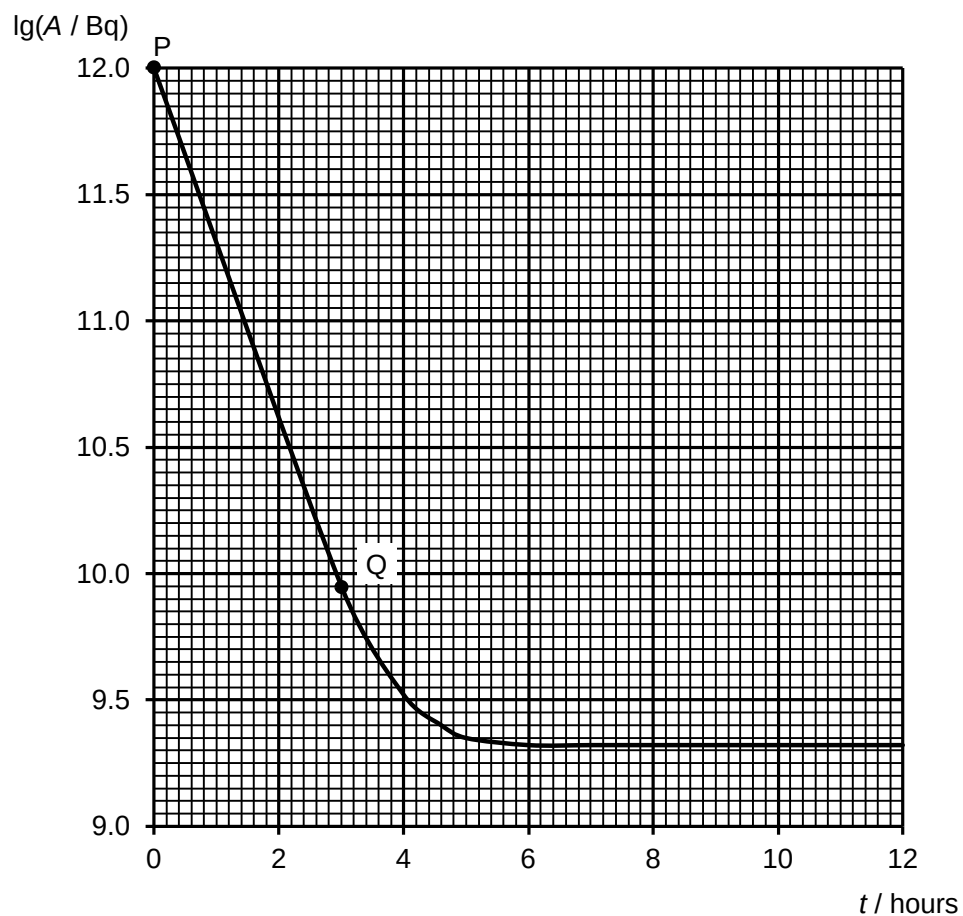


Fig. 7.1

- (i) Write down the nuclear equation for the decay of $^{131}_{53}\text{I}$ to Xe, showing the mass number and atomic number of the isotope Xe clearly.

..... [2]

- (ii) Explain why the part PQ on the graph corresponds mainly to the decay of $^{131}_{52}\text{Te}$.

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 [2]

(iii) Determine the gradient of PQ.

gradient = [2]

(iv) Hence, or otherwise,

1. show that the half-life of $^{131}_{52}\text{Te}$ is approximately 0.44 hours,

[3]

2. determine the initial number of $^{131}_{52}\text{Te}$ nuclides in the sample.

number of nuclides = [2]

(v) Explain why the graph seems flat after a time of 6 hours.

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 [2]

(vi) Hence, calculate the half-life of $^{131}_{53}\text{I}$.

half-life = hours [3]

(d) Suggest why background radiation can be ignored in this question.

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..... [1]

(e) State and explain whether these two radioactive nuclides would pose any hazard if found when de-commissioning a nuclear reactor.

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..... [2]

End of paper